Introduction to Module 5: Data Structures, Continued

Design and Analysis of Algorithms

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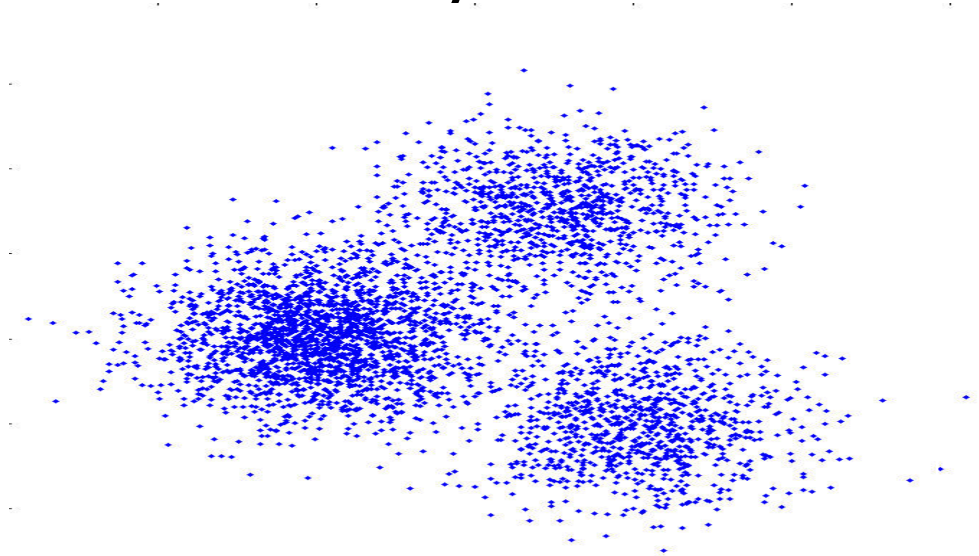
Main Topics

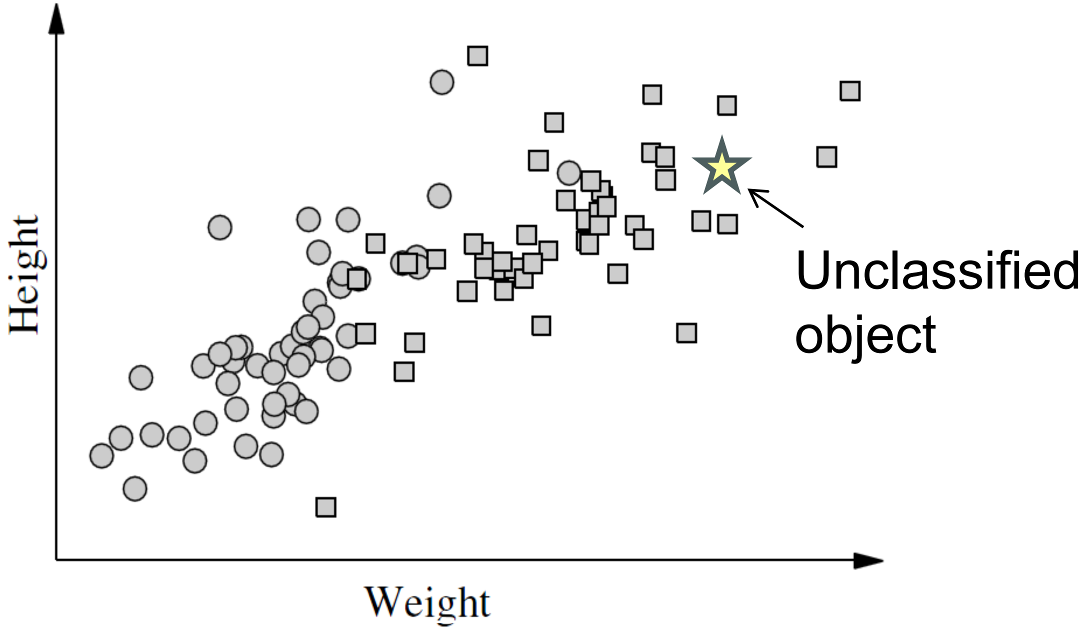
- Relevant for massive data sets / databases:
  - Range queries and updates
  - Multi-dimensional extensions of binary search trees
  - B-trees and their advantages in real-world memory systems
- Amortized analysis
- Review of data structure design methods
- Optimal enrichment lecture: Persistence
- Coding discussion: Multi-dimensional range queries

Coding Assignment: Fast k-Nearest Neighbor Queries

Sample applications:

- K-Nearest neighbor classification
- Measuring clustering / embedding consistency
- Dimensionality reduction





Notes on Course Content Moving Forward

- We’ve covered a lot of ground so far!
- We will cover a lot of ground in this and future modules also!
- Upcoming modules (this one included) will still cover a good number of fundamental topics, but also touch on more advanced material (often at just a high level).
- Please don’t be overwhelmed by the depth and breadth of material being covered!
  - Algorithmic problem solving skills improve by seeing examples.
  - In exams / assignments, we focus on mastery of fundamentals.